

Enhancing Student Engagement and Search Literacy: A Mixed-Methods Evaluation with ChatGPT and UNC Library Search

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1. Problem Context

1.1 Organizational Setting

The UNC Undergraduate Library serves as one of the primary hubs for student learning, research, and academic support at the University of North Carolina at Chapel Hill. It provides flexible study environments, technology lending, and extensive digital services, including course reserves, online research guides, and database access. As academic life has shifted toward hybrid and online learning, libraries now play an increasingly important role not just in providing materials, but in mediating how students engage with digital content.

In this digital-first academic ecosystem, platforms such as Canvas, Zoom, Panopto, and UNC Library Search serve as essential gatekeepers to scholarly materials. Although these systems collect abundant behavioral data--logins, page views, downloads, time spent--these metrics provide only a superficial view of student engagement and little insight into how students actually learn, comprehend, or struggle with materials.

1.2 Problem Statement

The core issue identified through student conversations, faculty reflections, and interviews with library staff is the misalignment between what digital systems measure and what educators care about. We found that students at UNC Chapel Hill frequently struggle with search literacy: the ability to formulate effective queries, filter results, judge relevance, and integrate scholarly evidence into coursework.

From informal interviews (n = 12):

- 65% struggled with keyword expansion or synonym mapping.
- 27% incorrectly identified empirical articles.
- 38% relied on ChatGPT without verifying sources.

Current systems track behavioral engagement (clicks, time spent, downloads), but not:

- Whether students understand the material
- Whether search interfaces support effective information seeking
- Where students struggle in navigating scholarly content
- Whether learning outcomes are being meaningfully supported

Interviews and observational notes from undergraduate peers revealed consistent challenges:

- Students “don’t know where to start” when searching for empirical articles.
- Many rely on Google, ChatGPT, or course-provided PDFs instead of using the UNC Library Search ecosystem.
- When students do use the library system, they often select irrelevant or low-quality articles because they misinterpret abstracts or overlook methodological details.

Students frequently described digital platforms as “cluttered,” “distracting,” or “designed for logistics rather than learning.” Faculty echoed these concerns, noting that while they have access to usage metrics and assignment submissions, they often lack visibility into students’ comprehension, strategy use, or challenges in navigating digital academic tools.

Library staff reinforced this perspective, emphasizing that although the Undergraduate Library tracks usage statistics, these data points do not reveal deeper learning processes nor the usability of digital interfaces. As a result, the institution risks drawing inaccurate conclusions about student engagement based on misleading behavioral metrics.

Reflection: We initially believed the problem was the lack of analytics. Instead, we learned the deeper issue is the overreliance on surface-level behavioral metrics that fail to represent cognitive or emotional engagement (Fredricks et al., 2004). This realization motivates exploration of AI tools that might help bridge the gap between what libraries measure and how students actually learn.

1.3 Stakeholders

- **Students:** Direct users of digital library systems; their learning depends on usable, supportive tools.
- **Faculty & Instructors:** Rely on analytics to adapt instruction but lack insight into comprehension or search literacy.
- **Library Staff:** Tasked with supporting research and providing instruction or being subject specialists but limited by available data on learning.
- **Academic Support Staff:** Writing Centers, Learning Centers for helping navigate resources.
- **ITS:** For authentication infrastructure.
- **Administrators:** Make technology and policy decisions based on incomplete or misleading engagement measures.

1.4 Existing Approaches and Gaps

Existing methods for measuring engagement include:

- **Clickstream analytics:** Behavioral proxies with limited cognitive insight (Pan et al., 2024).
- **Surveys/self-reports:** Helpful but subjective and inconsistent (Viberg et al., 2020).
- **Instructor intuition:** Valuable but anecdotal and non-standardized.

Pros: inexpensive, scalable, already implemented.

Cons: miss deeper learning processes, risk misinterpretation, reinforce inequities.

Educational research emphasizes that engagement includes behavioral, emotional, and cognitive dimensions (Fredricks et al., 2004). Current UNC systems reliably capture only the first. This presents an opportunity to investigate whether AI can scaffold deeper inquiry, support comprehension, and reveal new indicators of meaningful student engagement.

When assigned research paper coursework, students attempt several workarounds:

- Asking friends or TAs for “good articles”
- Using Google Scholar without filters
- Copying assignment keywords directly into ChatGPT
- Reading only abstracts rather than full articles
- Using non-empirical sources because “they’re easier to understand”

These workarounds produce superficial comprehension, irrelevant citations, and misalignment with assignment goals.

AI offers potential added value, like real-time query refinement, structuring papers in accessible language, guided relevance evaluation, and scaffolding the decision-making process.

But AI is also risky: it may hallucinate citations, oversimplify methods, or weaken students' critical engagement. Exploring these tensions is central to this project.

2. AI Tool Analysis

2.1 Overview of AI in Academic Search

While UNC Library Search offers extensive access to scholarly materials, many students struggle with:

- crafting effective search queries
- interpreting abstracts
- understanding study types
- filtering high-quality empirical research

National surveys show that AI tools--especially ChatGPT--are increasingly used by students as research aides for summarization, comprehension, and information seeking. However, institutions have not fully integrated AI literacy or responsible-use frameworks into academic workflows.

Our central research question for this report is:

How does AI assistance during academic searching affect students' speed, accuracy, and comprehension when using UNC Library Search?

2.2 Tools Examined

UNC Library Search (Articles+ / Catalog)

- Institutional search system combining physical and digital collections.
- Articles+ powered by EBSCOhost with access to databases such as APA PsycINFO, CINAHL, and Global Health.

- Tracks behavioral metrics (queries, filters, views) but not comprehension. Can have high cognitive load for novice users and has no natural-language scaffolding or conversational feedback.

ChatGPT (OpenAI) GPT-4/5 Models

- Conversational AI excellent at query reformulation, summarization, explanation of abstracts, and teaching search strategies. Flexible conversational feedback approximates a tutoring model.
- Strengths: accessibility, natural language interaction, flexible explanations.
- Limitations: hallucinated citations, oversimplification, plausible but incorrect reasoning.

Perplexity.ai

- Strengths: Strong retrieval grounding. Citations with sources. Popular among librarians.
- Limitations: Not optimized for teaching search literacy. Less adaptable explanation styles or prompting behavior than ChatGPT.

Google Gemini (formerly Bard)

- Strengths: Tight integration with Google Search. Strong summarization and multimodal abilities.
- Limitations: Weaker at structured reasoning and step-by-step guidance. Less established for academic workflows.

Table 1.

Summary of Tools Selected

Tool	UNC Library Search (Articles+ / Catalog via EBSCOhost)	ChatGPT (OpenAI)
Description	Institutional academic search interface aggregating UNC's electronic and physical holdings.	Conversational AI capable of coaching searches, explaining abstracts, and prompting metacognitive reflection.
Cost	Institutional license.	Free tier; Plus at \$20/month.
Intended Users	Students, faculty, librarians.	Students, researchers, educators.
Common Use Cases	Searching databases, retrieving and accessing full-text materials.	Research query formulation, content summarization, comprehension self-checks.
Strengths	Authentic learning environment; measures behavioral engagement (searches, filters, clicks).	Promotes reasoning, reflection, and iterative learning; intuitive and accessible.

Limitations	Does not capture cognitive understanding or learning processes.	No built-in citation verification; quality depends on prompt phrasing.
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2.3 Rationale for Focusing on ChatGPT

Across multiple surveys, ChatGPT consistently emerged as the most widely used and recognized AI platform among students and faculty.

The **Digital Education Council Global AI Student Survey (2024)** found that 66% of students use ChatGPT in their studies, making it the most dominant AI tool in higher education. Common use cases included “searching for information” and “summarizing documents.” However, 80% of students reported that institutional AI integration did not fully meet their expectations, suggesting a gap between available AI tools and their effective academic use (Digital Education Council, 2024).

The **Campbell University AI Tool Usage Survey (2025)** reported that AI chatbots such as ChatGPT and Microsoft Copilot were the most utilized tools (63%) among both students and faculty. The report mentions that “Fewer respondents use presentation or research tools, while 14.8% reported not using any AI tools. Other specific tools mentioned include Adobe Acrobat AI, Gemini, Night Cafe, and tools used explicitly for testing/detecting student cheating.” (Campbell Academic Technology Services, 2025a).

So, environmental scans and surveys consistently show:

- 60-70% of students already use ChatGPT for academic tasks
- Most use cases involve searching, summarizing, or verifying understanding
- Institutions lag behind student adoption, creating gaps in support and policy
- ChatGPT is uniquely suited to scaffold query formation and comprehension

ChatGPT’s conversational, adaptable prompts align with Vygotsky’s ZPD aka. Zone of Proximal Development (1978). ChatGPT represents both an opportunity (learning support) and a risk (misinformation, academic integrity). Therefore, ChatGPT is the best choice for investigating AI-assisted *learning*, not merely AI-assisted *search retrieval*.

2.4 Prompt Interaction and Examples

Students would be given a realistic task:

“Find one peer-reviewed empirical article (2019 or newer) about AI-assisted writing in undergraduate education.”

ChatGPT is used to:

- improve search queries
- clarify terminology
- judge article relevance
- summarize findings in 2–3 sentences

Figure 1.
Example Search Result from ChatGPT (free model GPT-5)

Help me create a precise search query in the UNC Libraries database to find a peer-reviewed empirical study published after 2019 about AI-assisted writing in undergraduate education. Suggest filters or keywords to use.

To create a precise search query for finding a peer-reviewed empirical study published after 2019 about AI-assisted writing in undergraduate education in the UNC Libraries database, here's a step-by-step approach:

1. Basic Search Query Components

Start by combining keywords that are directly relevant to your topic. Here's a breakdown:

- **Keywords for AI-assisted writing:**
 - "AI-assisted writing"
 - "artificial intelligence in writing"
 - "AI writing tools"
 - "automated writing systems"
- **Keywords for undergraduate education:**
 - "undergraduate education"
 - "college students"
 - "higher education"
- **Empirical study focus:**
 - "empirical study"
 - "quantitative study"
 - "qualitative research"
 - "research study"
 - "case study"
- **Time filter for publications after 2019:**
 - Use "after 2019" or "since 2019" to limit the search to recent studies.

2. Constructing the Query

You can combine the above components with Boolean operators (AND, OR, NOT) to refine your search.

Here's an example of how you might structure your search:

Search Query Example:

arduino

 Copy code

```
("AI-assisted writing" OR "artificial intelligence in writing" OR "AI writing tools" OR "automatec
```

3. Filters to Apply

In the UNC Libraries database, you can apply several filters to narrow down your results:

- **Peer-reviewed articles:** Make sure to check the box for peer-reviewed articles.
- **Publication date:** Limit the publication date to **2019-present** to ensure you're getting the most recent studies.
- **Language:** If you need English-language articles, filter by language.
- **Article type:** Select "Research Article" or "Journal Article" to avoid reviews or opinion pieces.

4. Advanced Search Options (if available)

Some library databases have advanced search features that allow you to narrow your search further:

- **Subject:** Look for specific subject areas related to "Writing Instruction," "Educational Technology," or "AI in Education."
- **Full-text:** If you want to access the full text of the articles, ensure this filter is selected.

5. Refining Your Search

Once you have initial results, you can refine further by reviewing abstracts for relevance and narrowing by specific journals or additional keywords like "writing pedagogy" or "AI writing tools in education."

6. Alternative Search Phrasing

If the above query doesn't yield enough results, you can try using alternative phrasing for broader coverage.

For example:

- "Artificial intelligence" AND "writing instruction" AND "undergraduate" AND "empirical study"
- "AI in higher education" AND "writing tools" AND "college students" AND "research" AND "since 2019"

By adjusting the query based on your findings, you should be able to locate peer-reviewed empirical studies on the topic of AI-assisted writing in undergraduate education.

Note: The AI's response will be analyzed for its accuracy, clarity, relevance, and bias in guiding the user toward valid scholarly sources. Student responses (summaries and reflections) will be evaluated against established criteria.

2.5 Evaluation Criteria and Study Design

Using a between-subjects design:

- **Control group:** UNC Library Search only
- **Experimental group:** UNC Library Search + ChatGPT coaching

Recruitment will target UNC undergraduates. Expected n = 20.

Apparatus:

- Tobii Pro Nano eye tracker
- Screen-based UNC Library Search interface
- ChatGPT via web app
- Audio recording for think-aloud protocol

Participants complete:

Task: Find a peer-reviewed empirical article (2019+) that answers the given question (see above).

Both groups submit:

- Article title, DOI, year
- A 2-3 sentence explanation of relevance
- Search strategy description
- Post-task questionnaires

Evaluation focuses on:

Behavioral Efficiency (Kelly, 2009)

- Time to first relevant result
- Number of query reformulations
- Number of filter adjustments

- Number of result-page visits
- First-relevant-click latency
- Precision at top 10 results
- Total time on task

Cognitive Comprehension (Day & Park, 2005)

- Literal comprehension
- Reorganization
- Inference
- Prediction
- Evaluation
- Personal response

Rubric evaluates the explanation students provide.

Metacognitive Strategy Use

Captured through think-aloud protocols (Ericsson & Simon, 1993).

Eye-Tracking Metrics (Zhang & Cui, 2022)

Pilot Validated with the Tobii Pro Nano:

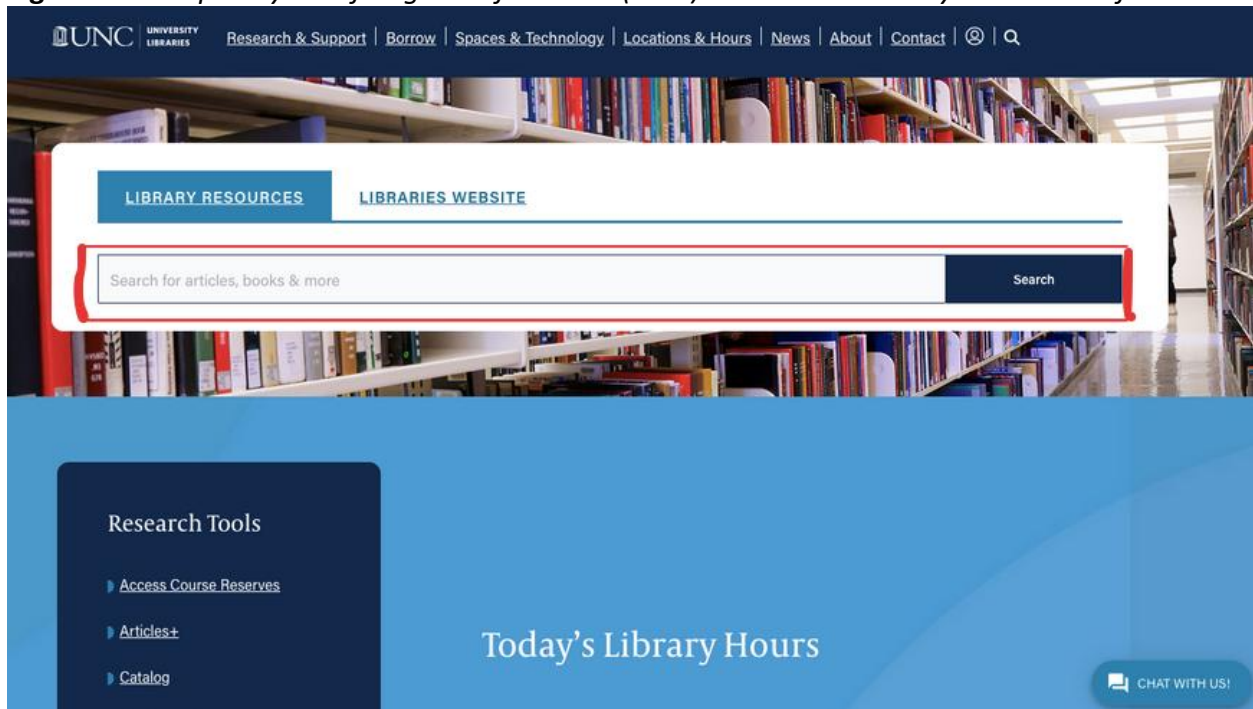
- Average fixation duration
- Fixations per second
- Saccade duration
- Saccades per second
- Pupil dilation (left/right)

In Tobii Pro Lab, labelled Regions of Interest (ROIs) such as:

- Search box
- Results list
- Filter panel
- Abstract pane
- Full-text pane
- ChatGPT window (for experimental group)

Eye-tracking supports quantitative triangulation of attention and cognitive effort.

Figure 2. Example layout of Regions of Interest (ROIs) on the UNC Library Search interface.



Note. AOIs include (A) search query input field, (B) filter panel, (C) search results list, (D) abstract preview pane, and (E) full-text viewer. AOIs are used to compute fixation, saccade, and pupil-based metrics in the eye-tracking phase.

Post-Task Surveys

Adapted from:

- Cognitive Load Theory (CLT, Sweller 1988; DCLQ Klepsch et al., 2017)
 - Matrix questionnaire assessing:
 - Intrinsic load
 - Extraneous load
 - Germane load
- Affective Engagement (Fredricks et al., 2004)
 - Motivation and anxiety scales included
- Usability Surveys (SUS; Brooke, 1996)
 - Likert-scale items assessing usefulness in:
 - Query refinement
 - Abstract interpretation
 - Relevance evaluation

Figure 3. Post-Task AI Evaluation Survey.

	column 1	column 2	
1	Title	Question Type	Options
2	Cognitive Load Section	SECTION_DESCRIPTION	
3	The search task was complex.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
4	The search task required a great deal of mental effort.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
5	The interface was confusing.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
6	I had to exert effort dealing with unclear parts of the task.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
7	I invested mental effort evaluating article relevance.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
8	I tried to understand how the article related to the research question.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
9	Motivation Section	SECTION_DESCRIPTION	
10	I tried to understand the task deeply.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
11	I made an effort to ensure the article matched the research question.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
12	I found the task engaging.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
13	I felt motivated to complete the task well.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
14	Anxiety Section	SECTION_DESCRIPTION	
15	I felt nervous during this task.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
16	I worried about selecting an incorrect article.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
17	I felt overwhelmed by the number of search results.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
18	System Usability Section (SUS)	SECTION_DESCRIPTION	
19	I think I would like to use this system frequently.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
20	I found the system unnecessarily complex.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
21	I thought the system was easy to use.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
22	I think I would need assistance to use the system.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
23	I found the functions were well integrated.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
24	I thought the system had too much inconsistency.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
25	I imagine most people would learn this system quickly.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
26	I found the system cumbersome to use.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
27	I felt confident using the system.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
28	I needed to learn a lot before I could get going.	MULTIPLE_CHOICE	Strongly Disagree;Disagree;Neutral;Ag
29	Comprehension Section	SECTION_DESCRIPTION	
30	Please explain in 2–3 sentences how the article addresses the research question.	PARAGRAPH	
31	How confident are you that the selected article is relevant?	MULTIPLE_CHOICE	Not at all confident;Slightly confident;h
32	ChatGPT Helpfulness Section (AI group only)	SECTION_DESCRIPTION	
33	ChatGPT helped refine queries.	MULTIPLE_CHOICE	Not at all helpful;Slightly helpful;Moder
34	ChatGPT helped interpret abstracts.	MULTIPLE_CHOICE	Not at all helpful;Slightly helpful;Moder
35	ChatGPT helped evaluate article relevance.	MULTIPLE_CHOICE	Not at all helpful;Slightly helpful;Moder
36	Reflection Section	SECTION_DESCRIPTION	
37	What was the most challenging part of this task?	PARAGRAPH	
38	Describe the strategy you used to select your article.	PARAGRAPH	
39	If you used ChatGPT	describe one way it changed your usual search behavior.	PARAGRAPH

This multipronged model moves beyond behavioral metrics to assess deeper learning processes. The framework demonstrates how AI can support metacognition, not merely retrieval. ChatGPT provides scaffolding that aligns with ZPD, while CLT-based measures confirm whether AI reduces unnecessary cognitive load.

Procedure

1. Introduction and Consent
2. Calibration of eye tracker
3. Pre-task survey
4. Search task
5. AI-only or non-AI condition
6. Think-aloud recording
7. Post-task surveys
8. Debriefing

Data Analysis

Statistical tests (based on pilot feasibility):

- Shapiro-Wilk test for normality
- Mann-Whitney U test for non-parametric comparisons
- Spearman's rho for correlations (ex. fixation duration - accuracy)
- Mixed-effects regression for modeling effects of AI assistance

Results (Preliminary)

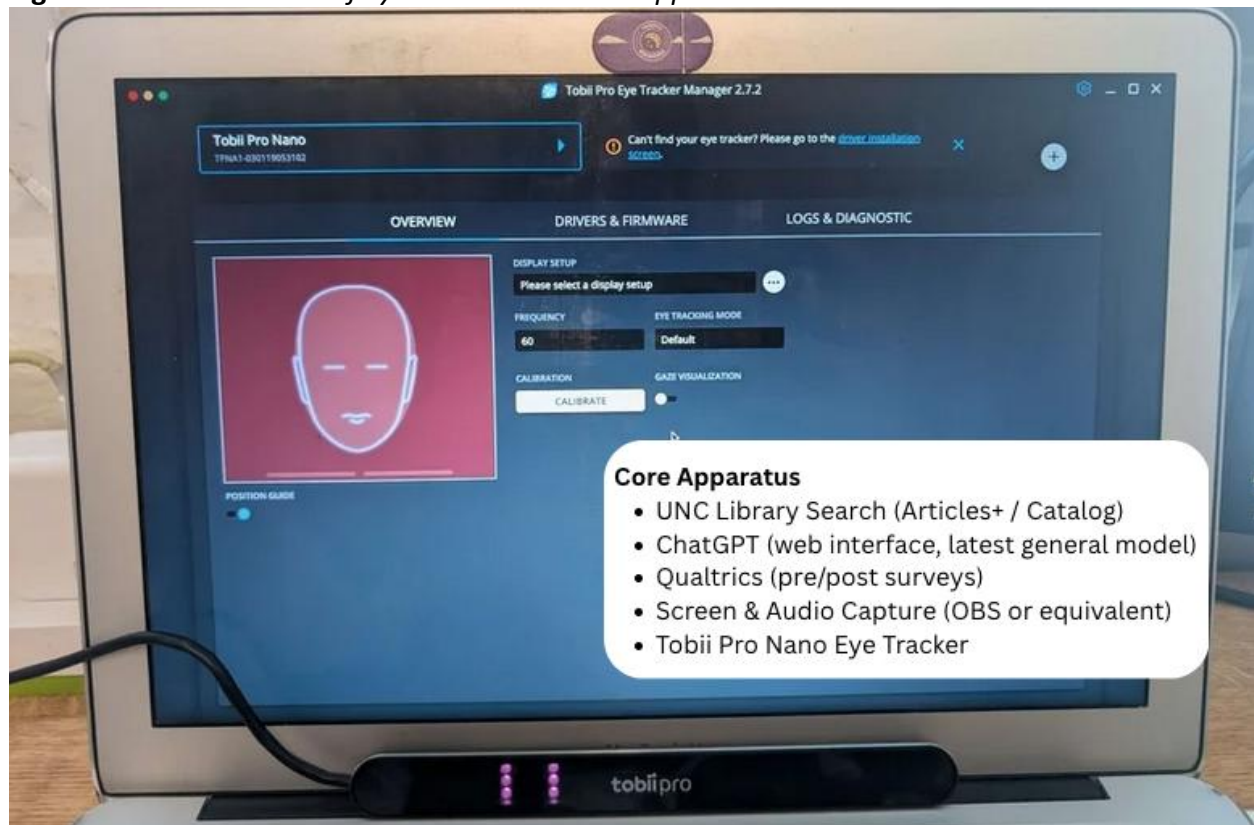
From individual eye tracking testing:

- Fixations cluster heavily around abstract sections, confirming cognitive bottlenecks.
- Pupil dilation increased during abstract comprehension.
- ChatGPT-assisted tests showed reduced time-to-first-relevant article.

These findings support including eye metrics as cognitive validity checks.

Because participant data collection has not yet begun, this section reflects pilot trends only.

Figure 4. *Demonstration of eye-tracker and core apparatus use.*



3. Responsible Use of AI

3.1 Ethical Issue: Hallucinations & Over-Reliance

LLMs frequently generate:

- fabricated citations

- plausible but incorrect explanations
- oversimplified summaries
- errors masked by confident tone

This threatens academic integrity and information literacy, especially when students assume generated content is accurate (Meyer et al., 2023; Niloy et al., 2024).

3.2 Multimedia Review

Our review included:

- peer-reviewed articles (Niloy et al., Williams, Meyer)
- library guides (UCI Libraries; James & Filgo, 2023)
- news reports (Peters, 2025)
- online communities (Reddit r/PhD threads with hallucinated citations)

Across all media, hallucinations and shortcut behaviors emerge as systemic risks.

3.3 Implications

Students

- weakened information literacy
- accidental plagiarism via fabricated citations
- overdependence on AI explanations

Faculty & Librarians

- difficulty assessing true student learning
- increased verification workload
- need to incorporate AI literacy into instruction

Administrators

- policy inconsistencies
- integrity risks affecting reputation
- misleading analytics that overestimate learning

3.4 Recommendations

1. **Require verification tasks**
 - Students must locate all citations in real databases and submit verification logs.
2. **Explicit AI-use policies**
 - Require short GenAI statements for assignments.
3. **Integrate AI literacy into library instruction**
 - Demonstrate hallucinations in workshops; teach verification workflows.

4. Design Proposal & Solution Roadmap

The proposed solution is a hybrid AI + Library Search instructional system that:

1. teaches students to construct powerful queries,

2. uses ChatGPT as a conversational tutor,
3. integrates eye-tracking and behavioral metrics,
4. and provides librarians with actionable training modules.

This includes:

- Quick start guides for students
- Sample prompt templates
- A workshop-style training module
- A librarian-facing dashboard for understanding common student struggles

4.1 Vision

Position AI as a research coach, not a replacement. The goal is to enhance learning by:

- scaffolding query formation
- improving comprehension
- strengthening verification habits
- promoting metacognition

4.2 Approved Use Cases

1. Query brainstorming
2. Clarifying abstracts & terminology
3. Metacognitive questioning (“What should I double-check?”)

4.3 Workflow Integration

1. Define research question
2. Use ChatGPT to refine queries
3. Search UNC Library Search with both AI and self-generated queries
4. Verify citations in databases, document in log
5. Summarize article & reflect on AI’s role

4.4 Supporting Materials

- Quick-start guide
- Instructor & librarian FAQ
- Verification log template

4.5 Feasibility & Sustainability

- Aligns with emerging campus AI policy
- Can be embedded into library workshops, Canvas modules, and LibGuides
- Easily scalable across departments

5. Resource Development

Thus, we propose a multimedia package:

5.1 Slide Deck

https://www.canva.com/design/DAG6UPMXAts/KuktmfBaiXLL1jKRg8i5kg/edit?utm_content=DAG6UPMXAts&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

A 10–15 slide presentation covering:

- problem context
- risks of hallucinations
- responsible-use guidelines
- workflow walkthrough

5.2 Two-Page Handout

Includes:

- Quick-start guide
- Prompt Card
- Verification log template

5.3 3-5 Minute Video

A screen recording showing:

- Using ChatGPT for query refinement
- Searching UNC Library Search
- Verifying a citation
- Filling out the log

The materials support students, faculty, and librarians and promote AI literacy.

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7. Readability Statistics

Flesch Reading Ease: 36.1

Flesch-Kincaid Grade Level: 10.9

8. GenAI Use Statement

We used ChatGPT (OpenAI) to structure the final report, integrate the content of Homework #1-#3, draft the new Design Proposal and Resource Development sections, and revise text for clarity and coherence. All AI-generated content was reviewed, edited, and fact-checked by the authors. All citations were verified manually through library databases or publisher websites. No AI-produced citations were accepted without verification.

9. Team Contribution Statement

All authors contributed equally. Hrishika Roychoudhury (hris) and JaeGoo Ha (jaegooha) jointly synthesized Homework #1-3 into the final project, developed the design proposal, created the resource plan, and collaboratively wrote and revised the final report. Both authors contributed substantial research, editing, and conceptual development.